Final report of a Mach-Zehnder Interferometer design

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***Abstract*—This document describes the design and test of a Mach-Zehnder interferometer realized in Silicon-on-Insulator photonics. We report the results obtained from the electromagnetic simulation that was performed using Lumerical MODE, INTERCONNECT and Matlab. The design is laid out on KLayout. The purpose of the study is to understand the behavior of the MZI across different lengths. Parameters of interest are: the effective index, goup index, and the free space range (FSR) as functions of wavelengths.**

***Index Terms*—Mach-Zehnder Interferometer, Silicon Photonics, Optical Interconnects.**

I. Introduction

In silicon photonics optical interconnect, the Mach-Zehnder interferometers (MZIs) are widely used to control the phase and amplitude of light signals for applications such as optical switches, modulators, and wavelength division multiplexers (WDMs). Mach-Zehnder interferometers (MZIs) work by splitting a light beam into two paths, or "arms," and then recombining them to create an interference pattern. The interference pattern can be manipulated by introducing a phase shift in one of the arms.

In this project a phase shift is introduced by changing the physical lengths of the two beams. A path length difference changes the phase of the light wave, as different parts of the wave are recombined. To achieve a controlled phase shift, the length difference must be less than the coherence length of the light source, a distance over which its waves maintain a predictable phase relationship. In this article we propose a Mach-Zehnder inteferometer with two fiber graded couplers and provide its simulation results when their length difference, ΔL, varies.

II. Theory

An imbalanced MZI physical properties with different path length is tabulated below. A laser input is fed into the inteferometer through a sub-wavelength coupler, split by a Y-branch to a pair of waveguides of different length that act as phase shifters, and recombined at the other end, to be observed externally at the output by yet another coupler.

A. Compact model for the waveguide in polynomial expression

The compact model for the waveguide can be expressed in polynomial terms for $\lambda_0 = 1.55um$.

$$n_{eff} = 2.45 - 1.13(\lambda - 1.55) - 0.04(\lambda - 1.55)^2 \quad (1)$$

We use Matlab to calculate the n_{eff} across wavelength from 1.5 to 1.6 nm using equation 1, and the group index and dispersion can be derived respectively by using the following analytical expressions.

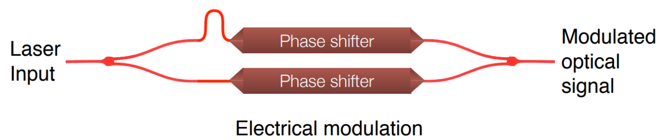


Fig. 1: An imbalanced Mach-Zehnder Interferometer[1]

TABLE I: Physical properties of the MZI

Waveguide Type	Strip
Waveguide thickness, width	220nm, 500nm
Polarization	Quasi-TE
Interferometer imbalance length, ΔL	75, 100, 150, 200 um
Interferometer type	Mach-Zehnder Interferometer
Splitter type	Y-branch
Design variations	Free space spacing (FSR in nm) vs. ΔL, group delay vs. width

$$n_g = n_1 - n_2\lambda \quad (2)$$

$$D = -\frac{\lambda}{c}2n_3 \quad (3)$$

where, from equation 1, $n_1 = 2.45, n_2 = -1.13, n_3 = -0.04$ and $c = 200,000\,km/sec$, the speed of light in the air.

B. Equation for the interferometer - Transmission vs. Wavelength

It can be shown[1] that the relationship between the intensity level at the output and input of an imbalanced interferometer waveguides with identical propagation constant β, where it is 100 % fully constructive when $\Delta L = 0$, and fully destructive when $\Delta L = \frac{\pi}{\beta}$.

$$\frac{I_o}{I_i} = \frac{1}{2}[1 + \cos(\beta \Delta L)] \quad (4)$$

C. Free Spectral Range vs. Wavelength

In this project we will vary ΔL in 75, 100, 150, 200um. The free spectral range (FSR) of the interferometer can be analytically found by the following formula,

$$FSR(\lambda) = \lambda^2/(n_g \Delta L) \quad (5)$$

The values of the table II is calculated based on the formula given in the equation 5 above and will be compared against simulations, and eventually against the final real design.

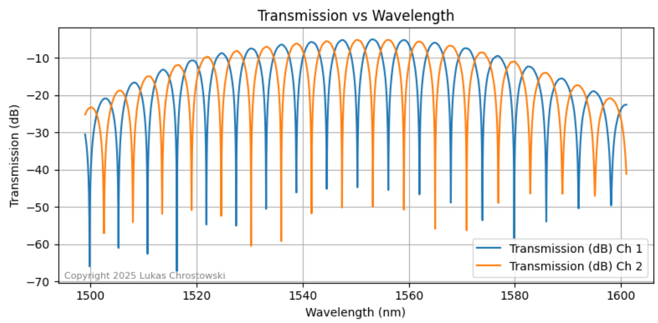


Fig. 2: Transmission (dB) vs wavelength at $\Delta L = 100nm$ for 2 different channels, as shown in Figure 3.

TABLE II: Analytical $FSR(\lambda)$ vs. ΔL

$\Delta L(um)$	$FSR(nm)$
75	7.620
100	5.715
150	3.810
200	2.8577

III. Modeling and Simulation

A. Lumerical Finite Difference Eigenmode Solver

First, mode simulation is done on the Lumerical Finite Difference Eigenmode (FDE) Solver. We need to find the spatial field distribution of the TE and TM modes for ifferent waveguide dimensions.

Figure 4 shows the Electric-field intensity profile in TE mode of a 500nm x 220 nm waveguide. This is simulated from the FDE mode.

Figure 5 shows the Magnetic-field intensity profile in TM mode of the same 500nm x 220 nm waveguide. In both profiles, it is shown that all intensities are confined within the waveguide nicely.

The effective index and group index across wavelengths are also profiles in Figure 6 and 7 for the TE and TM modes respectively.

B. MZI Simulation with Lumerical INTERCONNECT schematic

Simulation with $\Delta L = 200$ has been completed, shown in Figure 10. The design of the Mach-Zehnder Interferometer with fiber graded couplers have been simulated on Lumerical INTERCONNECT, and a transmission plot vs wavelength is shown in Figure 11 with and without the couplers. Without the couplers, the plot is flat without any insertion loss.

C. MZI Simulation with Lumerical INTERCONNECT directly from KLayout

Figure 12, 13, 14, and 15 illustrates simulation of the transmission in dB vs wavelength in nm for various $\Delta L = (75, 100, 150, 200)\,um$. It can be observed from the plots that the free spectral range (FSR), i.e. the peak-to-peak distance on the curve from the plots, and tabulated in Table III, agree with the analytical calculation given in Table II.

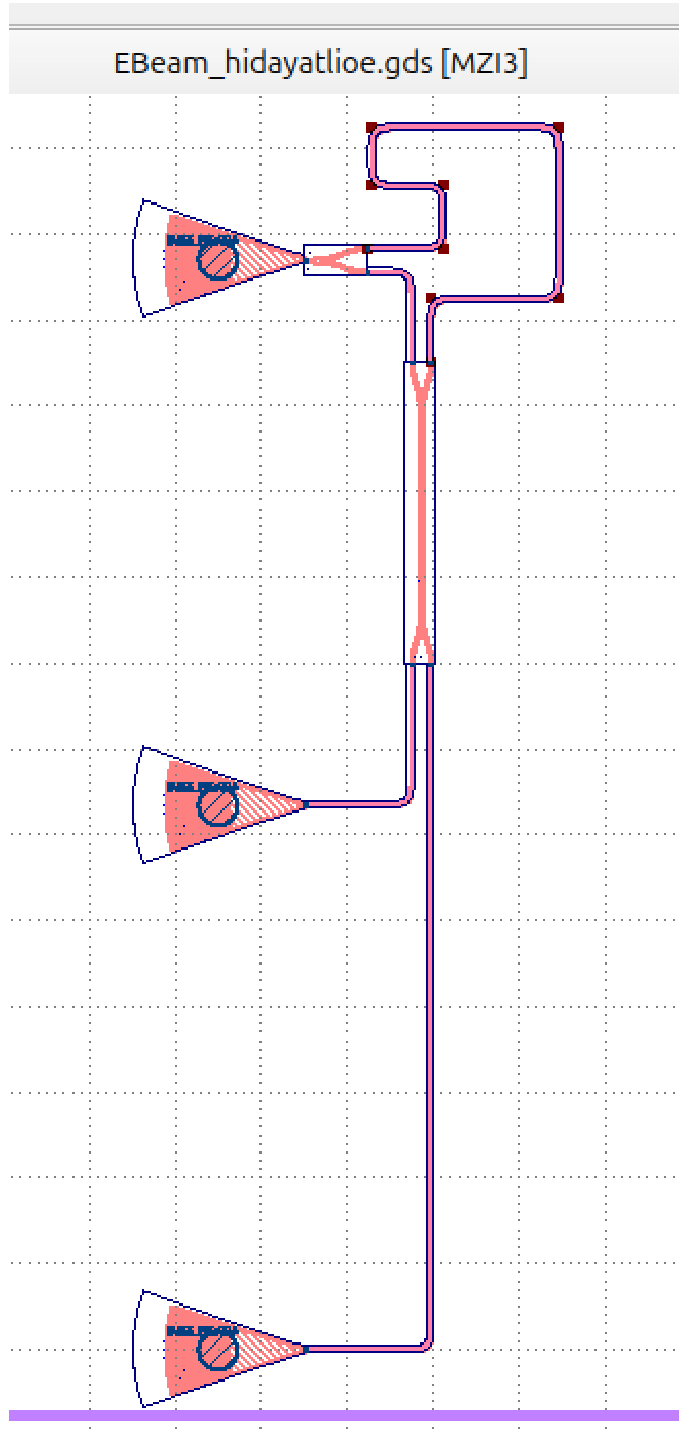


Fig. 3: KLayout of the MZI with one input and two different output channels.

As a reference, a single waveguide with similar physical characteristic is designed and simulated, and its transmission (dB) vs. wavelength (nm) is shown in Figure 18a without any phase shift interference.

IV. Fabrication[3]

Corner analysis for 9 different points have been simulated. This is used to anticipate variations during fabrication pro-

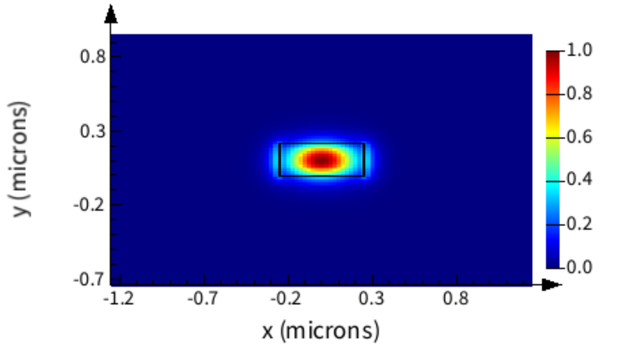


Fig. 4: TE Mode E-Intensity profile of a 500 nm x 220 nm waveguide with Lumerical's FDE[3]

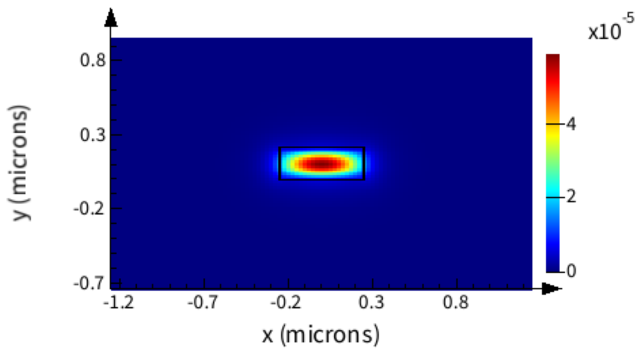


Fig. 5: TM Mode H-Intensity profile of a 500 nm x 220 nm waveguide with Lumerical's FDE[3]

cesses which can result in structural mismatches.

As reported by Washington University Nanofabrication Facility, there is a change of ±20 nm in waveguides width. Beside, silicon thickness of the used wafer is not exactly 220 nm, and varies from 203 to 212 nm.

It can be shown from the plot in Figure 8a and Figure 8b for the TE and TM modes respectively that the effective index and group index varies across corners, and the effective index can vary from 2.34 to 2.47 at the intended wavelength of 1550nm.

V. Summary of the Fabrication

Fabricated is performed at one or more of these: Applied Nanotools and Washington Nanofabrication Facility. The following are the process descriptions.

Applied Nanotools, Inc. NanoSOI process: The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H_2SO_4:H_2O_2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectometry measurements were performed throughout the

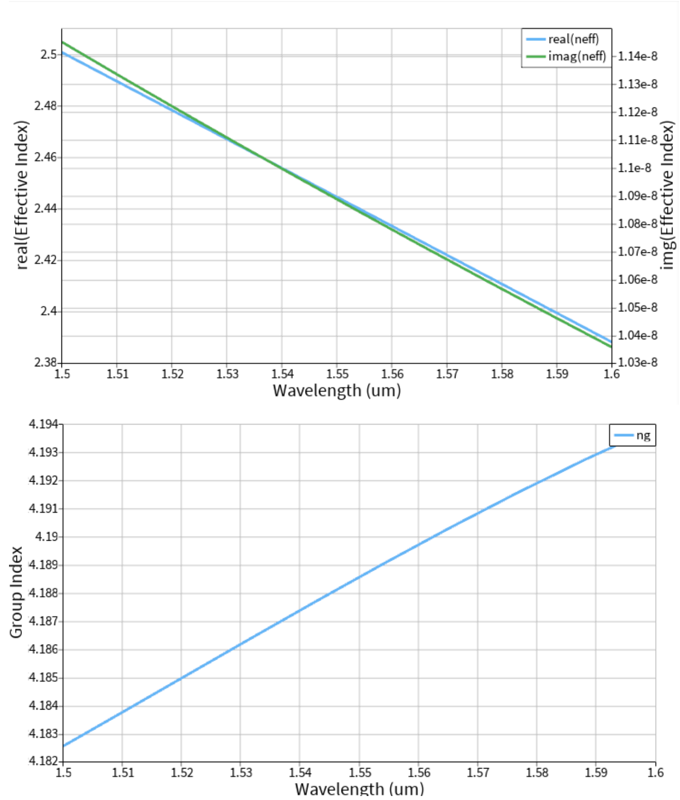


Fig. 6: Effective index vs. wavelength and group index vs. wavelength of the 500 nm x 220 nm in TE mode

TABLE III: Simulated $FSR(\lambda)$ vs. ΔL

$\Delta L(um)$	$FSR(nm)$
75	7.550
100	5.720
150	3.850
200	2.870

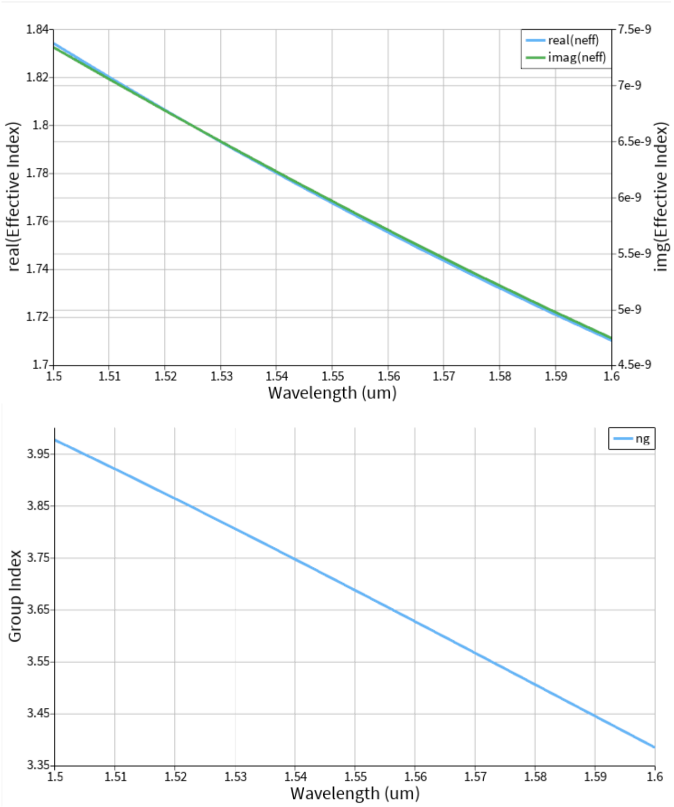
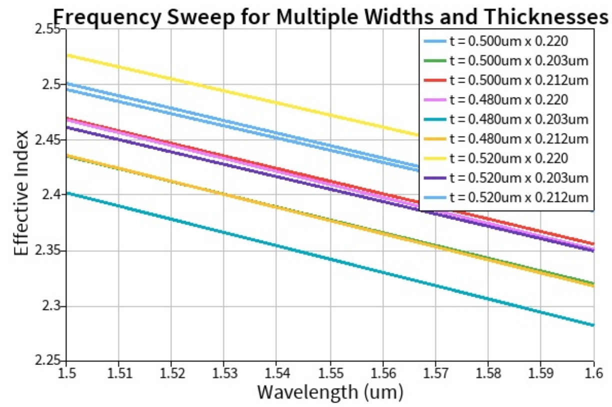
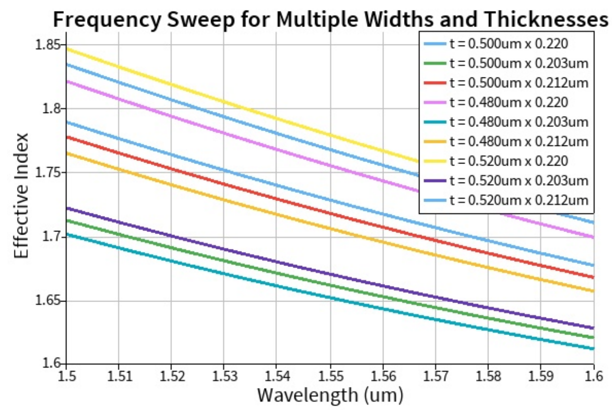


Fig. 7: Effective index vs. wavelength and group index vs. wavelength of the 500 nm x 220 nm in TM mode



(a) 9 corners for the TE modes



(b) 9 corners for the TM modes

Fig. 8: Corner analysis of 9 corners for the TE and TM modes

process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

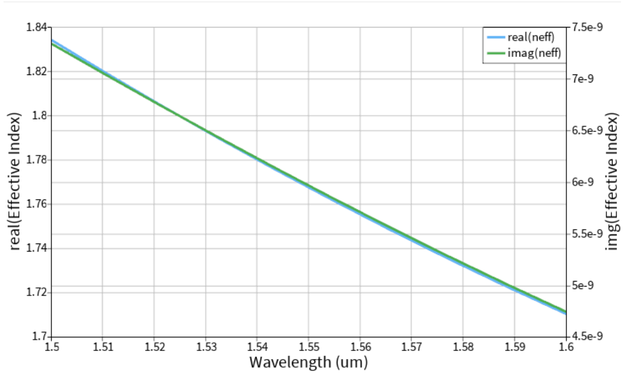
Washington Nanofabrication Facility (WNF) silicon photonics process The devices were fabricated using 100 keV Electron Beam Lithography [1]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 um thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 °C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 µm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 uC/cm2 was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using plasma enhanced chemical vapor deposition (PECVD) in an Oxford Plasmalab System 100 with a silane (SiH4) flow of 13.0 sccm, nitrous oxide (N2O) flow of 1000.0 sccm, high-purity nitrogen (N2) flow of 500.0 sccm, pressure at 1400mT, high-frequency RF power of 120W, and a platen temperature of 350C. During deposition, chips rest directly on a silicon carrier wafer and are buffered by silicon pieces on all sides to aid uniformity.

VI. Measurement description

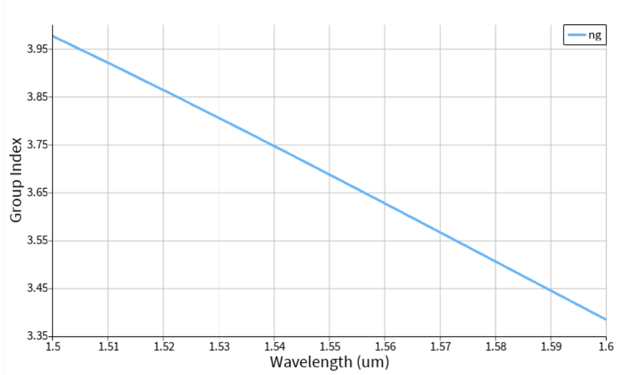
To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90º rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

VII. Data Analysis

Indeed we have received the result data from our different MZI designs with different delta lengths.



(a) Effective index of the TE mode



(b) Effective index of the TM mode

Fig. 9: Effective index vs. wavelength of the TE and TM modes

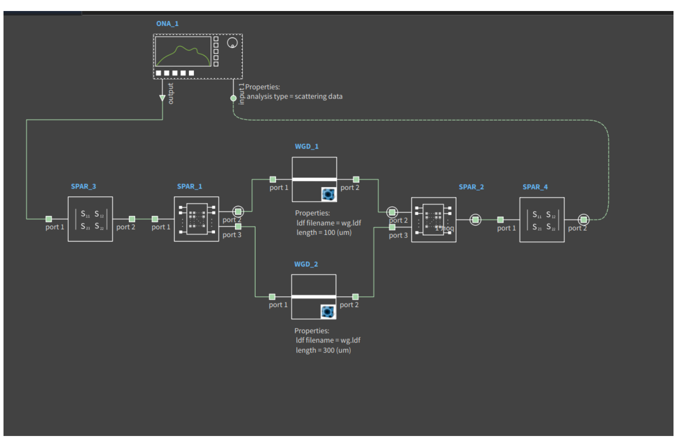


Fig. 10: Mach-Zehnder Interferometer design with Lumerical's INTERCONNECT[1]

The transmission characteristics of several data in dB are plotted in Figure 16, Figure 17, and Figure 18 for different delta lengths.

We saw insertion losses of approximately 15 dB of final fabricated data as compared to the simulation data.

VIII. Conclusion and Ackowledgement

We acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Omid Esmaeeli performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

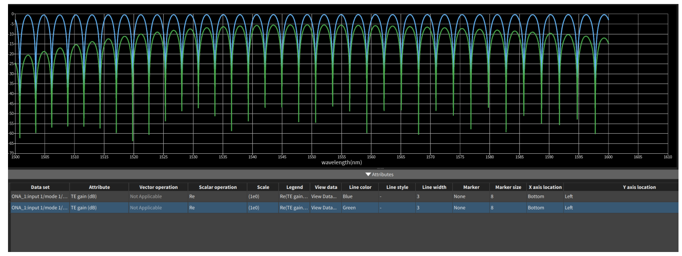


Fig. 11: Insertion Loss of the Mach-Zehnder Interferometer with and without couplers[1]

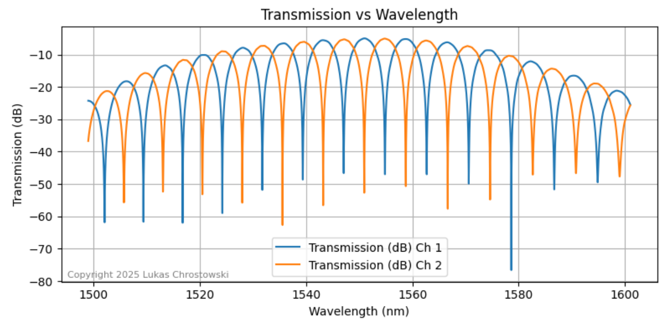


Fig. 12: Transmission (dB) vs wavelength at $\Delta L = 75um$, as observed FSR=7.55nm

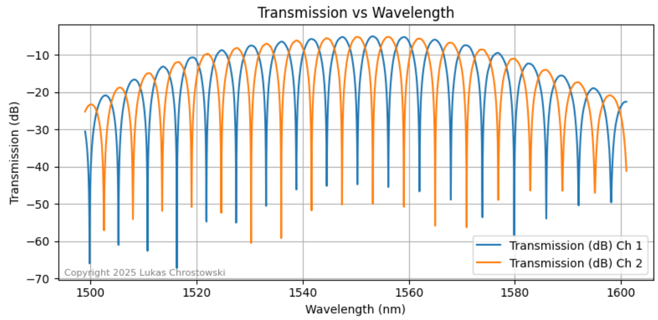


Fig. 13: Transmission (dB) vs wavelength at $\Delta L = 100um$, as observed FSR=5.72nm

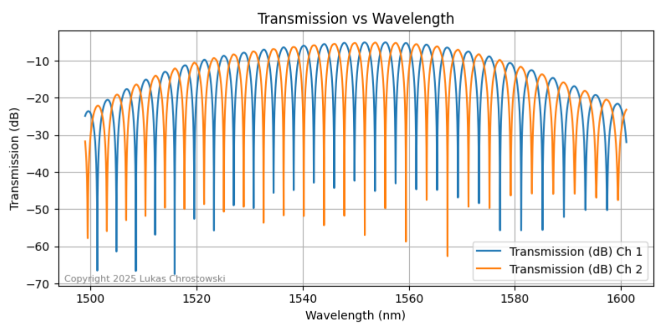


Fig. 14: Transmission (dB) vs wavelength at $\Delta L = 150um$, as observed FSR=3.85nm

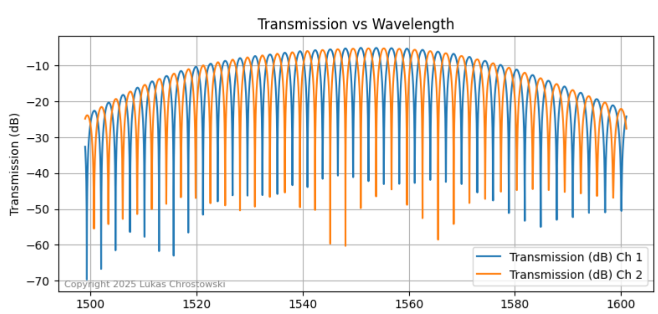
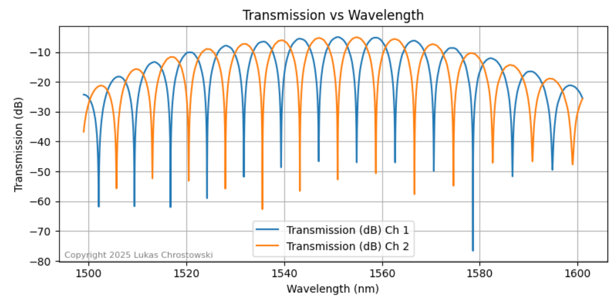
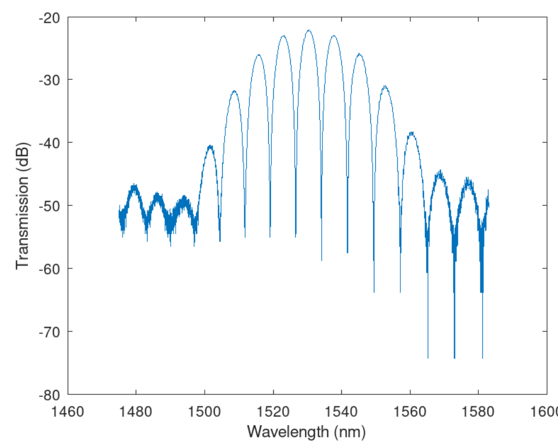


Fig. 15: Transmission (dB) vs wavelength at $\Delta L = 200um$, as observed FSR=2.87nm

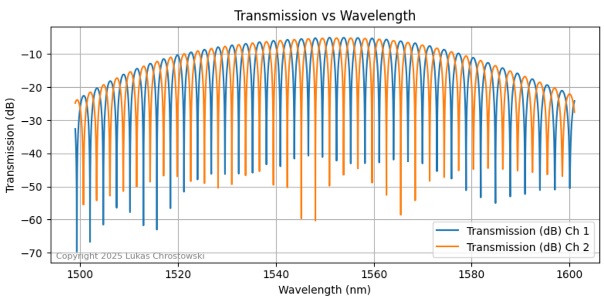


(a) Transmission (dB) vs wavelength in a simulated MZI design

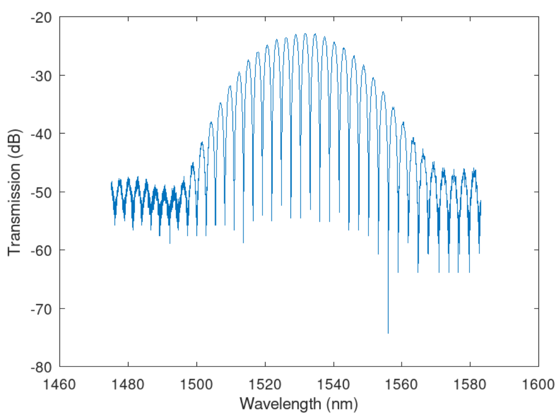


(b) Transmission (dB) vs wavelength in the final fabricated MZI design

Fig. 16: Comparison between (a) simulated MZI, and (b) fabricated MZI design delta L=75um, with FSR=7.55nm. We saw an insertion loss of approximately 15 dB of the final design as compared to simulation.

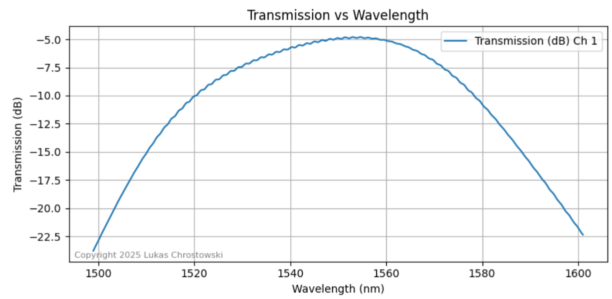


(a) Transmission (dB) vs wavelength in a simulated MZI design

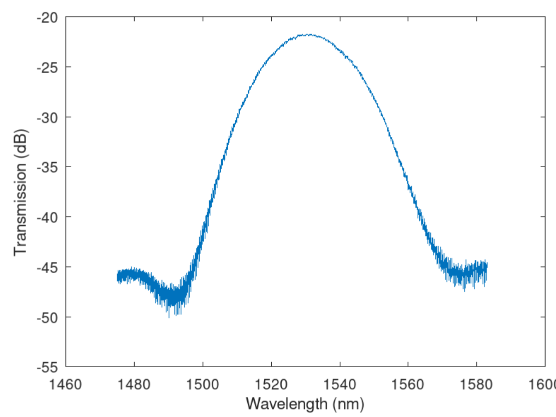


(b) Transmission (dB) vs wavelength in the final fabricated MZI design

Fig. 17: Comparison between (a) simulated MZI, and (b) fabricated MZI design delta L=200um, with FSR = 2.87nm. We saw an insertion loss of approximately 15 dB of the final design as compared to simulation.



(a) Transmission (dB) vs wavelength in a simulated MZI design



(b) Transmission (dB) vs wavelength in the final fabricated MZI design

Fig. 18: Comparison between (a) simulated MZI, and (b) fabricated MZI design. We saw an insertion loss of approximately 15 dB of the final design as compared to simulation.

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